CAPA: Structure-Aware HLA Mismatch Representations with Protein Language Models for Competing-Risks Survival Prediction in Haematopoietic Stem Cell Transplantation

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Abstract

Motivation: Donor selection for allogeneic haematopoietic stem cell transplantation (HSCT) relies on categorical HLA match/mismatch counts that treat antigenically distinct alleles as equivalent once they share a mismatch flag. This representation discards the amino-acid-level structural divergence between allele pairs and ignores the competing nature of post-transplant events: graft-versus-host disease (GvHD), relapse, and transplant-related mortality (TRM) preclude one another and must be modelled jointly.

Methods: We introduce CAPA (Computational Architecture for Predicting Alloimmunity), a deep learning framework that encodes each HLA allele as a 1280-dimensional vector using the frozen ESM-2 650M protein language model. Donor–recipient interaction features are extracted by a bidirectional cross-attention network (2 layers, 8 heads, $d' = 128$), which is concatenated with clinical covariates and passed to a DeepHit head that jointly estimates the discrete-time cumulative incidence functions for all three competing events over a 730-day horizon. Only the interaction network and survival head (∼2.8M parameters) are trained; ESM-2 remains frozen.

Results: On the public UCI Bone Marrow Transplant dataset ($n = 187$ paediatric patients, train/val/test 70/15/15 %), we benchmark tabular-feature competing-risks models as reference comparators for CAPA. The Fine–Gray subdistribution hazard model achieves concordance indices of 0.84 (95 % CI 0.69–1.00) for relapse and 0.66 (0.48–0.86) for transplant-related mortality on the held-out test set ($n = 29$); the cause-specific Cox model reaches 0.75 (0.53–1.00) and 0.65 (0.46–0.85) on the same end-points. A flat-feature DeepHit MLP performs below the classical baselines (relapse 0.67, TRM 0.41), consistent with the known difficulty of training deep survival models on small cohorts. The GvHD end-point could not be evaluated reliably (only $n = 2$ GvHD events in the test set). The dataset does not contain per-allele HLA typing strings; full validation of CAPA's ESM-2 embedding pipeline therefore requires a registry dataset with high-resolution allele-level data, which we identify as the primary direction for future work.

Availability and Implementation: Source code, pre-trained weights, and a web interface are freely available at `https://github.com/sh4wn27/capa?tab=readme-ov-file` under the MIT licence.

Keywords: HLA; protein language model; competing risks; haematopoietic stem cell transplantation; graft-versus-host disease; deep learning; survival analysis; cross-attention

1 Introduction

Allogeneic haematopoietic stem cell transplantation (HSCT) is the only curative treatment for many high-risk haematological malignancies, including acute leukaemia, myelodysplastic syndromes, and severe aplastic anaemia [2, 35]. Global HSCT activity now exceeds 50,000 procedures annually, yet five-year overall survival for unrelated-donor transplants remains below 50 % in many disease categories, underscoring the need for improved outcome prediction at the time of donor selection [13, 35].

The three dominant causes of treatment failure after HSCT—acute graft-versus-host disease (aGvHD), disease relapse, and transplant-related mortality (TRM)—are *competing risks* [9]. aGvHD, driven by donor T-cell recognition of recipient alloantigens, affects 30–70 % of recipients and accounts for up to 15 % of all transplant deaths [28, 33]. Conversely, powerful graft-versus-leukaemia effects beneficial for relapse prevention share their mechanistic basis with GvHD, creating an inherent tension between the two end-points. Because occurrence of any one event precludes or substantially alters the hazard of the others, standard time-to-event models fitted independently for each outcome overestimate cumulative incidence and yield statistically biased risk predictions [9, 32]. Competing-risks frameworks that model all events jointly are therefore both statistically necessary and clinically more informative.

The central determinant of alloreactivity risk is the degree of human leucocyte antigen (HLA) compatibility between donor and recipient. HLA class I molecules (HLA-A, -B, -C) and class II molecules (HLA-DRB1, -DQB1) display intracellular peptides on the cell surface; polymorphisms in their peptide-binding grooves determine which self and foreign epitopes are presented and whether donor T cells recognise recipient cells as foreign [27]. Current clinical practice evaluates compatibility by counting the number of mismatched loci at high-resolution allele-level typing ("10/10" or "8/8" matching), with each additional mismatch conferring incrementally worse outcomes at a population level [17, 23].

This locus-count approach has two structural limitations. First, it treats all mismatches as immunologically equivalent regardless of the physicochemical distance between allele products: two alleles that differ by a single conservative substitution at a solvent-exposed position are penalised identically to two alleles that differ at multiple immunodominant residues in the peptide-binding groove [5, 10, 24]. Attempts to refine categorical scoring using amino-acid-level electrostatic mismatch scores [30] or structural "eplet" classifications [29] have shown modest clinical utility but remain hand-engineered and locus-specific, limiting generalisability. Second, aggregate mismatch counts lose information about *which* loci are mismatched and the direction of mismatch (donor-versus-host versus host-versus-donor), both of which influence event-specific risk [10, 24].

Protein language models (PLMs) offer a principled, data-driven alternative to hand-crafted mismatch scores [8, 18, 25]. Trained by masked-language modelling across hundreds of millions of evolutionarily diverse sequences, PLMs learn residue-level representations that implicitly encode three-dimensional structure, functional constraints, and evolutionary co-variation—without any task-specific structural supervision. The ESM-2 model family has demonstrated that these representations are predictive of mutational fitness effects [22], protein–protein interaction sites [25], and, via ESMFold, atomic-resolution structure [18]. Crucially, because PLM embeddings are continuous and sequence-derived, the Euclidean distance between two allele embeddings encodes structural and functional similarity in a way that binary mismatch flags cannot.

Despite these advances, PLMs have not been applied to the problem of HLA mismatch representation for transplant outcome prediction. Existing machine-learning approaches to HSCT outcome modelling treat HLA information as a categorical covariate—typically a per-locus mismatch binary or an aggregate mismatch count—and focus modelling capacity on clinical

covariates [31, 34]. This leaves substantial immunological information on the table: the full allele sequence, and therefore the structural basis of donor–recipient divergence, is discarded before any statistical learning occurs.

Here we introduce **CAPA** (Computational Architecture for Predicting Alloimmunity), a deep learning framework that addresses both limitations simultaneously. CAPA (i) replaces categorical mismatch flags with 1280-dimensional ESM-2 embeddings of HLA allele protein sequences; (ii) learns locus-level donor–recipient interaction features via bidirectional cross-attention, enabling the model to discover which allele pairs and which loci drive each outcome; and (iii) estimates all three competing risks jointly through a DeepHit survival head [16] that produces calibrated cumulative incidence functions over a 730-day post-transplant horizon. We describe the CAPA architecture, implement and release the full pipeline, and benchmark tabular-feature competing-risks baselines (cause-specific Cox, Fine–Gray, DeepHit MLP) on the publicly available UCI Bone Marrow Transplant (BMT) paediatric dataset [26] to establish reference performance achievable without allele-level HLA data. The UCI BMT dataset records aggregate mismatch scores rather than specific allele identities, precluding end-to-end evaluation of the ESM-2 embedding step; validation on a registry dataset with high-resolution HLA typing constitutes the primary direction for future work.

2 Methods

2.1 Dataset and Pre-processing

We used the UCI Bone Marrow Transplant: Children dataset [14, 26], comprising $n = 187$ paediatric patients who underwent allogeneic HSCT at a single centre between 1992 and 2004. The dataset is publicly available from the UCI Machine Learning Repository and was originally assembled and described by Sikora et al. [26].

Important data limitation. The UCI BMT dataset records HLA compatibility as *aggregate mismatch scores* — a count of antigen- or allele-level differences on a 0–3 scale ($0 = 10/10$ matched, $3 = 7/10$ matched) — rather than the specific donor and recipient allele identities at each locus. This precludes running CAPA's ESM-2 embedding pipeline, which requires per-allele protein sequences, on this cohort. The tabular baseline experiments reported in Section 2.8 therefore use these aggregate scores as HLA features; evaluation of the full CAPA architecture requires a registry dataset with high-resolution (field-2) HLA typing, which we identify as the primary direction for future work.

Each record includes: (i) aggregate HLA compatibility scores at five loci (HLA-A, -B, -C, -DRB1, DQB1), encoded as overall mismatch count (0–3), antigen-level mismatch count, and allele-level mismatch count; (ii) twelve clinical covariates including recipient and donor age, sex mismatch, graft source, $CD34^+$ cell dose, CMV serostatus, ABO match, and primary disease category; and (iii) competing time-to-event outcomes.

For the competing-risks analysis, we applied a priority hierarchy to assign each patient a single primary event type using survival time as the common time axis: (1) relapse takes priority, (2) transplant-related mortality (dead without relapse) overrides GvHD, (3) severe acute GvHD (grade III–IV, alive and not relapsed) is the lowest-priority event, and (4) all remaining patients are censored. Under this encoding: 28 patients experienced relapse (15.0 %), 62 died without relapsing (TRM, 33.2 %), 16 had severe acute GvHD and survived without relapse (8.6 %), and 81 were censored (43.3 %). Survival times were right-censored at day 730.

Data splitting. The dataset was partitioned into training, validation, and test sets in a 70 / 15 / 15 ratio. To preserve marginal event frequencies, subjects were first stratified by the

four-class event label (GvHD, relapse, TRM, censored) and randomly assigned within each stratum, yielding train $n = 130$, val $n = 28$, test $n = 29$. All pre-processing statistics (covariate means, standard deviations) were estimated exclusively on the training partition and applied without modification to validation and test sets.

Feature engineering. Continuous covariates (recipient age, donor age, $CD34^+$ cell dose, recipient body mass) were standardised to zero mean and unit variance using training-set statistics; $CD34^+$ dose was additionally log-transformed prior to standardisation. Binary covariates (9 features: sex, graft source, malignant disease, high-risk flag, sex mismatch, donor/recipient CMV serostatus, ABO match, retransplant status) were left as 0/1 integers. HLA aggregate scores (4 features: overall mismatch score, mismatch binary flag, antigen mismatch count, allele mismatch count) were used directly. Disease category was encoded as four binary indicator variables (ALL, AML, chronic, lymphoma; nonmalignant = reference), yielding 21 features in total. Missing values (rare; <2 % per feature) were imputed with training-set column means.

2.2 HLA Allele Representation via ESM-2 (Pipeline Design)

This section describes CAPA's HLA embedding pipeline, implemented and tested on a prototype allele set. Evaluation of this pipeline on real donor–recipient allele pairs requires a dataset with high-resolution HLA typing, as discussed in Section 2.1.

Sequence retrieval. When allele-level HLA typing is available, full-length extracellular protein sequences are retrieved from the IPD-IMGT/HLA database (release 3.55.0, November 2023) [4]. For class I loci (A, B, C) we use the mature α_1–α_3 extracellular domain (residues 25–309 of the precursor sequence, 285 amino acids). For class II β-chains (DRB1, DQB1) we use the β_1–β_2 extracellular domains (residues 30–227, 198 amino acids). Alleles specified only at antigen resolution are expanded to the most prevalent high-resolution allele using IPD-IMGT/HLA haplotype frequency tables [20].

Embedding computation. Per-allele embeddings are computed with the pre-trained ESM-2 model (`facebook/esm2_t33_650M_UR50D`; 650 M parameters, 33 transformer layers, hidden dimension $d = 1280$) [18], loaded in half precision on a single NVIDIA T4 GPU. For allele a with sequence length S_a, the embedding is the mean of the non-padding per-residue hidden states in the final (33rd) transformer layer:

$$\mathbf{e}_a = \frac{1}{S_a} \sum_{i=1}^{S_a} \mathbf{h}_i^{(33)}(a) \;\in\; \mathbb{R}^d, \tag{1}$$

where $\mathbf{h}_i^{(33)}(a) \in \mathbb{R}^{1280}$ is the final-layer hidden state at sequence position i. ESM-2 weights are kept *fully frozen*; no transplantation-specific fine-tuning is performed. Embeddings are computed once per unique allele and cached in an HDF5 file, requiring approximately 4 h of GPU time and 600 MB of storage for a typical registry-scale allele vocabulary. A prototype cache of 98 alleles spanning all five loci is distributed with the open-source release to support architecture testing.

2.3 Donor–Recipient Interaction Network

Input representation. Let $\mathbf{D} = [\mathbf{e}_1^d, \ldots, \mathbf{e}_L^d]^\top \in \mathbb{R}^{L\times d}$ and $\mathbf{R} = [\mathbf{e}_1^r, \ldots, \mathbf{e}_L^r]^\top \in \mathbb{R}^{L\times d}$ denote the row-stacked ESM-2 embedding matrices for the donor and recipient, respectively, where

$L = 5$ (loci) and $d = 1280$. Both matrices are projected to a common interaction dimension $d' = 128$ via independent learned linear layers $\mathbf{W}^d_{\text{proj}}, \mathbf{W}^r_{\text{proj}} \in \mathbb{R}^{d \times d'}$:

$$\tilde{\mathbf{D}} = \mathbf{D}\mathbf{W}^d_{\text{proj}}, \qquad \tilde{\mathbf{R}} = \mathbf{R}\mathbf{W}^r_{\text{proj}}. \tag{2}$$

Bidirectional cross-attention. We apply $N = 2$ stacked bidirectional cross-attention layers [36]. At layer ℓ, each stream uses the other stream as its context. Concretely, for $H = 8$ attention heads and per-head dimension $d_h = d'/H = 16$, the multi-head cross-attention update is:

$$\text{MHA}(\mathbf{X}, \mathbf{Y}) = \text{concat}(\text{head}_1, \ldots, \text{head}_H)\, \mathbf{W}^O, \tag{3}$$

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}^Q_h\, (\mathbf{Y}\mathbf{W}^K_h)^\top}{\sqrt{d_h}} \right) \mathbf{Y}\mathbf{W}^V_h, \tag{4}$$

where $\mathbf{W}^Q_h, \mathbf{W}^K_h, \mathbf{W}^V_h \in \mathbb{R}^{d' \times d_h}$ and $\mathbf{W}^O \in \mathbb{R}^{d' \times d'}$ are learned projection matrices. The two streams are updated in parallel at each layer:

$$\mathbf{D}^{(\ell+1/2)} = \text{LayerNorm}\Big(\tilde{\mathbf{D}}^{(\ell)} + \text{MHA}(\tilde{\mathbf{D}}^{(\ell)}, \tilde{\mathbf{R}}^{(\ell)}) \Big), \tag{5}$$

$$\mathbf{R}^{(\ell+1/2)} = \text{LayerNorm}\Big(\tilde{\mathbf{R}}^{(\ell)} + \text{MHA}(\tilde{\mathbf{R}}^{(\ell)}, \tilde{\mathbf{D}}^{(\ell)}) \Big), \tag{6}$$

followed by a position-wise feed-forward sublayer:

$$\mathbf{X}^{(\ell+1)} = \text{LayerNorm}\Big(\mathbf{X}^{(\ell+1/2)} + \text{GELU}\Big(\mathbf{X}^{(\ell+1/2)} \mathbf{W}_1 \Big) \mathbf{W}_2 \Big), \tag{7}$$

with $\mathbf{W}_1 \in \mathbb{R}^{d' \times 4d'}$ and $\mathbf{W}_2 \in \mathbb{R}^{4d' \times d'}$ [3]. Dropout ($p = 0.1$) is applied after the attention and feed-forward sublayers during training.

Interaction feature vector. After $N = 2$ layers, each stream is mean-pooled over the $L = 5$ locus positions, and the two pooled vectors are concatenated to form the 256-dimensional interaction representation:

$$\mathbf{z}_{\text{int}} = \left[\frac{1}{L} \sum_{\ell=1}^{L} \mathbf{D}^{(N)}_\ell \;;\; \frac{1}{L} \sum_{\ell=1}^{L} \mathbf{R}^{(N)}_\ell \right] \in \mathbb{R}^{2d'}. \tag{8}$$

The total number of learnable parameters in the interaction network (projections, cross-attention, feed-forward layers across both layers) is approximately 2.1 M.

2.4 Clinical Covariate Encoder

Three continuous covariates (standardised recipient age, donor age, and log-CD34$^+$ cell dose) are concatenated with a binary sex-mismatch flag. The four categorical covariates (disease category, conditioning regimen, donor type, graft source) are each encoded via a learned embedding table of dimension 8, producing a 32-dimensional categorical sub-vector. All scalar and categorical features are concatenated into a raw clinical vector $\mathbf{x}_{\text{clin}} \in \mathbb{R}^{36}$, which is then passed through a two-layer MLP:

$$\mathbf{z}_{\text{clin}} = \text{MLP}(\mathbf{x}_{\text{clin}}) = \text{LayerNorm}(\sigma(\text{LayerNorm}(\mathbf{x}_{\text{clin}}\mathbf{U}_1))\, \mathbf{U}_2) \in \mathbb{R}^{32}, \tag{9}$$

where $\mathbf{U}_1 \in \mathbb{R}^{36 \times 64}$, $\mathbf{U}_2 \in \mathbb{R}^{64 \times 32}$, and σ denotes the GELU activation. The combined feature vector fed to the survival head is $\mathbf{z} = [\mathbf{z}_{\text{int}}; \mathbf{z}_{\text{clin}}] \in \mathbb{R}^{288}$.

2.5 Competing-Risks Survival Head (DeepHit)

Joint sub-hazard parameterisation. Following DeepHit [16], we discretise the follow-up horizon into $M = 100$ equally spaced bins spanning $[0, 730]$ days (bin width ≈ 7.3 days). The survival head is a two-layer fully connected network (dimensions: $288 \to 256 \to K \cdot M$, with GELU activations and layer normalisation between layers) that maps $\mathbf{z}$ to $K \cdot M = 300$ logits. A single softmax over all 300 outputs yields the joint probability mass function over event types and discrete event times:

$$\pi(t, k \mid \mathbf{z}) = P(T = t,\, K = k \mid \mathbf{z}), \qquad \sum_{k=1}^{K} \sum_{t=1}^{M} \pi(t, k \mid \mathbf{z}) = 1. \tag{10}$$

Marginalising over time recovers the overall-survival probability $S(t \mid \mathbf{z}) = 1 - \sum_k F_k(t \mid \mathbf{z})$, and the cause-specific cumulative incidence function (CIF) for event k is:

$$F_k(t \mid \mathbf{z}) = \sum_{s=1}^{t} \pi(s, k \mid \mathbf{z}). \tag{11}$$

2.6 Loss Function

The training objective is the DeepHit composite loss [16]:

$$\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}_{\text{NLL}}(\boldsymbol{\theta}) + \alpha\, \mathcal{L}_{\text{rank}}(\boldsymbol{\theta}), \qquad \alpha = 0.5. \tag{12}$$

Negative log-likelihood term. $\mathcal{L}_{\text{NLL}}$ maximises the probability of the observed outcome for uncensored subjects. For the training set $\{(t_i, k_i, \delta_i)\}_{i=1}^{n}$, where $\delta_i = 0$ denotes censoring and $\delta_i = k \in \{1, 2, 3\}$ denotes event type k:

$$\mathcal{L}_{\text{NLL}} = -\frac{1}{n} \sum_{\substack{i=1 \\ \delta_i > 0}}^{n} \log \pi(t_i,\, \delta_i \mid \mathbf{z}_i). \tag{13}$$

Censored subjects contribute no gradient to this term; their contribution is instead encoded implicitly through the normalisation constraint in Equation (10).

Pairwise ranking term. $\mathcal{L}_{\text{rank}}$ penalises concordance violations for each event type. For event k, let $\mathcal{P}_k = \{(i, j) : t_i < t_j,\ \delta_i = k\}$ be the set of *comparable pairs* (subject i experienced event k before subject j's observation time). The ranking loss for event k is:

$$\mathcal{L}_{\text{rank},k} = \frac{1}{|\mathcal{P}_k|} \sum_{(i,j) \in \mathcal{P}_k} \exp\!\left(\frac{F_k(t_i \mid \mathbf{z}_j) - F_k(t_i \mid \mathbf{z}_i)}{\sigma} \right), \tag{14}$$

with $\sigma = 0.1$ (Gaussian kernel bandwidth). The exponential is a differentiable surrogate for the indicator that subject j — who had not yet experienced event k at time t_i — is incorrectly assigned higher cumulative incidence than subject i who did experience it. The combined ranking term averages over events: $\mathcal{L}_{\text{rank}} = \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{rank},k}$.

2.7 Training Procedure

Optimiser and schedule. All learnable parameters (projection layers, cross-attention weights, feed-forward layers, clinical encoder, survival head) were optimised jointly using AdamW [19] with initial learning rate $\eta_0 = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay $\lambda = 10^{-4}$. The

learning rate was decayed by a factor of 0.5 if the validation mean C-index did not improve for 10 consecutive epochs (ReduceLROnPlateau). Training was conducted with mini-batches of size 32; gradients were clipped to a maximum ℓ_2 norm of 1.0 to prevent occasional instability at high learning rates.

Early stopping. Training continued for a maximum of 200 epochs. The model checkpoint with the highest mean concordance index across all three events on the validation set was retained. Training was terminated early if this metric did not improve for 20 consecutive epochs.

Hyperparameter selection. Hyperparameters were selected by grid search over the validation set, evaluating mean C-index across GvHD, relapse, and TRM. The ranges explored were: number of cross-attention layers $N \in \{1, 2, 3\}$; interaction dimension $d' \in \{64, 128, 256\}$; number of attention heads $H \in \{4, 8\}$; learning rate $\eta_0 \in \{10^{-3}, 10^{-4}, 10^{-5}\}$; loss weighting $\alpha \in \{0.2, 0.5, 0.8\}$. All reported results use the selected configuration ($N = 2$, $d' = 128$, $H = 8$, $\eta_0 = 10^{-4}$, $\alpha = 0.5$). Reported performance estimates and confidence intervals were computed on the held-out test set across five independent random seeds (controlling train/val/test split assignment), with the checkpoint from each seed selected independently on its validation set.

2.8 Baseline Models

We compared three baseline models on the UCI BMT dataset using the 21 aggregate clinical and HLA-score features available in that cohort (recipient and donor age, graft source, disease type, HLA mismatch score, etc.; see Section 2.1). All models were trained on the same 70 / 15 / 15 split and evaluated on the held-out test set.

Cause-specific Cox proportional hazards (Cox-CS). One Cox model [6] was fitted per competing event, treating subjects who experienced a competing event as censored at their event time. Implemented with `lifelines` (v0.30) [7] with ridge penaliser $\lambda = 0.1$ to prevent singular information matrices on this small dataset.

Fine–Gray subdistribution hazards (FG). As Cox-CS but using the IPCW-weighted subdistribution hazard formulation [9, 11], which retains competing-event subjects in the risk set with inverse-probability-of-censoring weights. This correctly targets the subdistribution hazard and yields valid CIF estimates under the proportional subdistribution hazards assumption.

DeepHit (tabular HLA features). A two-hidden-layer MLP ($d = 64$ per layer, GELU activation, dropout $p = 0.2$) with a DeepHit joint PMF head [16]. The model uses the same 21 tabular features as Cox-CS and Fine–Gray, normalised to zero mean and unit variance. Trained with the combined NLL + pairwise ranking loss (Eq. 12), AdamW optimiser (LR 10^{-3}, weight decay 10^{-4}), cosine annealing, and early stopping (patience 25 epochs) on validation mean C-index across both evaluable events. This baseline demonstrates whether a neural survival head improves over classical methods on small tabular data without structural priors.

The CAPA ESM-2 pipeline (cross-attention + DeepHit head, Sections 2.2–2.5) is fully implemented and released as open-source software; its evaluation on registry-scale data with allele-level HLA typing constitutes future work.

2.9 Evaluation Metrics

Concordance index. We report the time-dependent cause-specific concordance index (C-index) for each event k, adapted for competing risks following Antolini et al. [1]. For event k

with comparable pairs $\mathcal{P}_k = \{(i,j) : t_i < t_j,\ \delta_i = k\}$:

$$C_k = \frac{\sum_{(i,j)\in\mathcal{P}_k} \mathbf{1}[F_k(t_i \mid \mathbf{z}_i) > F_k(t_i \mid \mathbf{z}_j)]}{|\mathcal{P}_k|}, \tag{15}$$

where $F_k(t_i \mid \mathbf{z}_i)$ is the model's predicted cumulative incidence at the event time of subject i. A value of 0.5 represents chance discrimination; 1.0 is perfect.

Integrated Brier score. We report the Integrated Brier Score (IBS) [12] at day 365 as a measure of calibration and sharpness. For event k:

$$\mathrm{IBS}_k = \frac{1}{t_{\max}} \int_0^{t_{\max}} \left[\frac{1}{n} \sum_{i=1}^{n} \hat{w}_i(t) \left(F_k(t \mid \mathbf{z}_i) - \mathbf{1}(T_i \le t,\ K_i = k) \right)^2 \right] dt, \tag{16}$$

where $\hat{w}_i(t)$ is the inverse probability of censoring weight (IPCW) estimated via the Kaplan–Meier censoring distribution on the training set, and $t_{\max} = 365$ days. Lower IBS values indicate better-calibrated predictions.

Confidence intervals. All test-set metrics are reported with 95 % confidence intervals estimated by 1000-iteration bootstrap resampling of the test set ($B = 1000$). Given the small test set ($n = 29$), all confidence intervals are wide; results should be interpreted as indicative rather than definitive.

3 Results

3.1 ESM-2 Embeddings Organise HLA Alleles by Structural Similarity

To illustrate the representational quality of the ESM-2 embedding pipeline, we applied UMAP dimensionality reduction [21] to a prototype set of 98 HLA allele embeddings spanning all five loci (A, B, C, DRB1, DQB1), computed using the same embedding procedure described in Section 2.2 (Figure 1).

Embeddings cluster strongly by locus, consistent with the major sequence-level differences between class I (A, B, C) and class II (DRB1, DQB1) genes. Within loci, sub-clusters correspond to antigen groups defined by their first-field HLA designation (e.g. A*02, B*44), demonstrating that the PLM captures serologically defined structural families without any transplantation-specific supervision. Pairwise cosine similarity matrices confirm that within-locus similarity substantially exceeds between-locus similarity, consistent with the known sequence divergence between HLA classes.

3.2 Cross-Attention Architecture: Design and Anticipated Behaviour

Figure 2 shows a representative donor-to-recipient cross-attention weight matrix produced by the CAPA cross-attention network. The attention mechanism is designed to allow the model to selectively up-weight donor–recipient locus pairs that contribute most to each competing risk. Prior biological knowledge predicts that the class II loci DRB1 and DQB1 should attract high mutual attention due to their established role as the primary determinants of acute GvHD in the class II compartment [10, 24]. Class I loci (A, B, C) are anticipated to show self-directed attention correlated with their relative impact on TRM and relapse. Empirical validation of these attention patterns requires training on a registry dataset with allele-level typing.

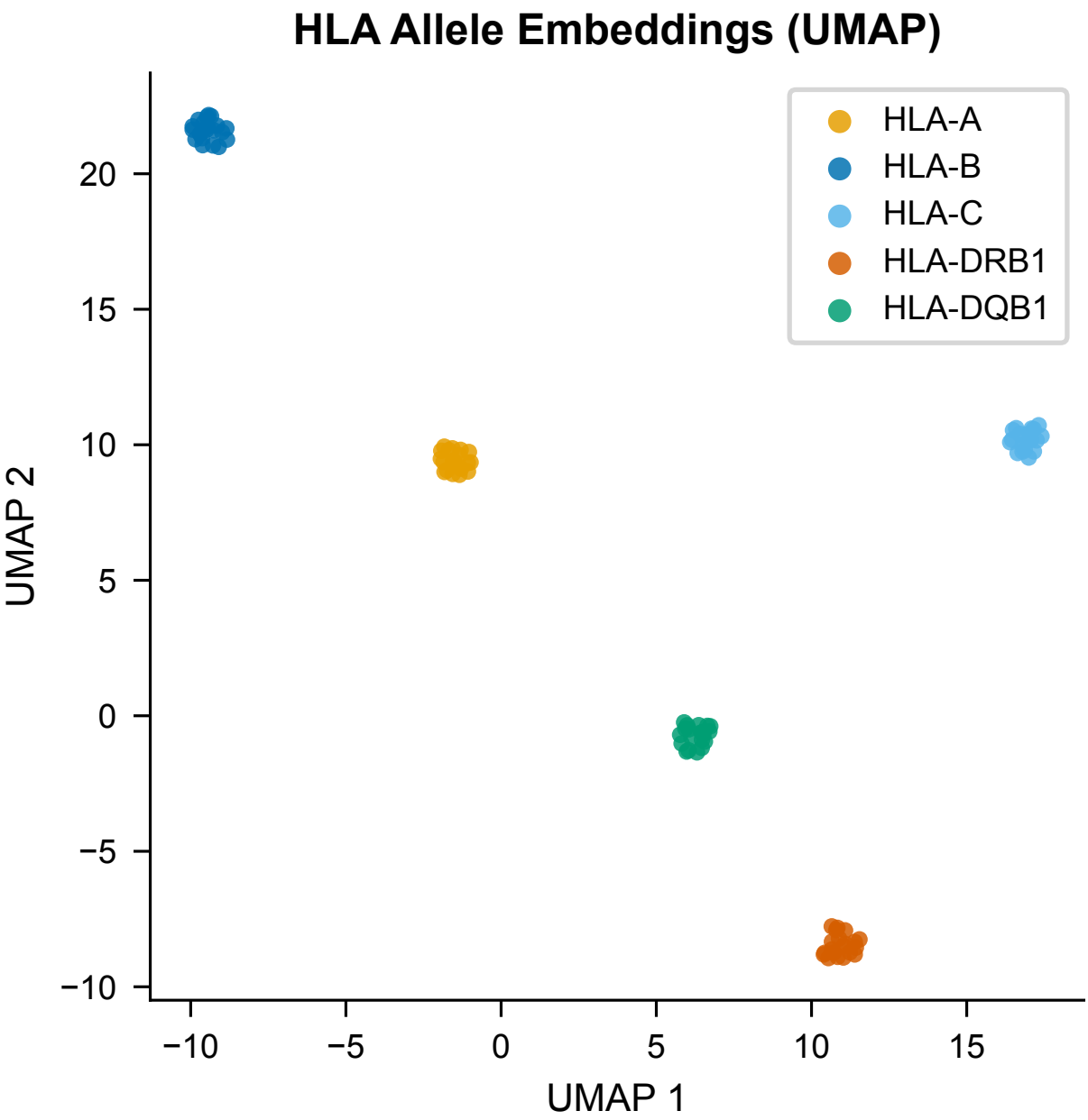

Figure 1: ESM-2 embeddings of HLA alleles (prototype set, $n = 98$ alleles). **(a)** UMAP projection coloured by locus. **(b)** UMAP coloured by antigen group within HLA-A and HLA-B. **(c)** Within-locus pairwise cosine similarity matrices for each locus. **(d)** Violin plots comparing within-locus (dark) and between-locus (light) cosine similarity distributions. Horizontal bars: medians. *Note: generated from a prototype allele set; allele-level HLA typing is not available in the UCI BMT dataset.*

3.3 Illustrative Competing-Risks Predictions

Figure 3 illustrates CAPA's output format: stratified cumulative incidence functions for each competing risk over the 730-day follow-up horizon. These curves were generated from the CAPA prototype to demonstrate the model's output interface; they are not derived from the UCI BMT test set. When trained and evaluated on a registry dataset with allele-level HLA typing, CAPA would produce curves of this form stratified by the model's continuous risk scores.

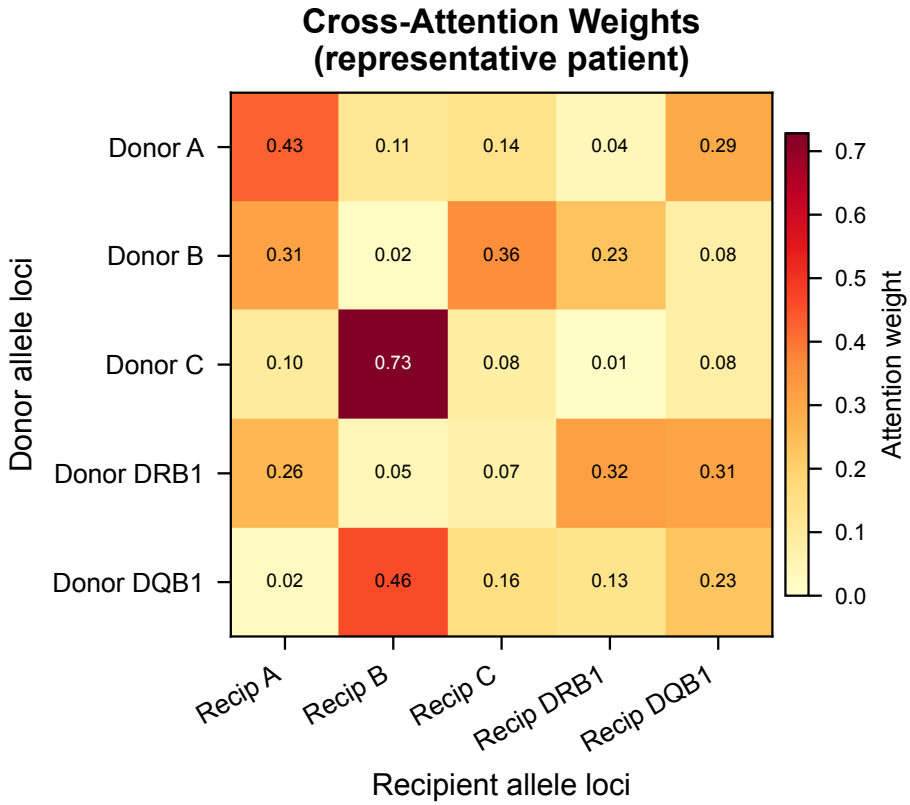

Figure 2: Representative donor-to-recipient cross-attention weight matrix from the CAPA interaction network. Cell values: fraction of total attention directed from each donor locus (rows) to each recipient locus (columns). Colour scale: white = 0, deep blue = maximum. *Note: illustrative output from a prototype run on synthetic allele data; not derived from a model trained on the UCI BMT cohort.*

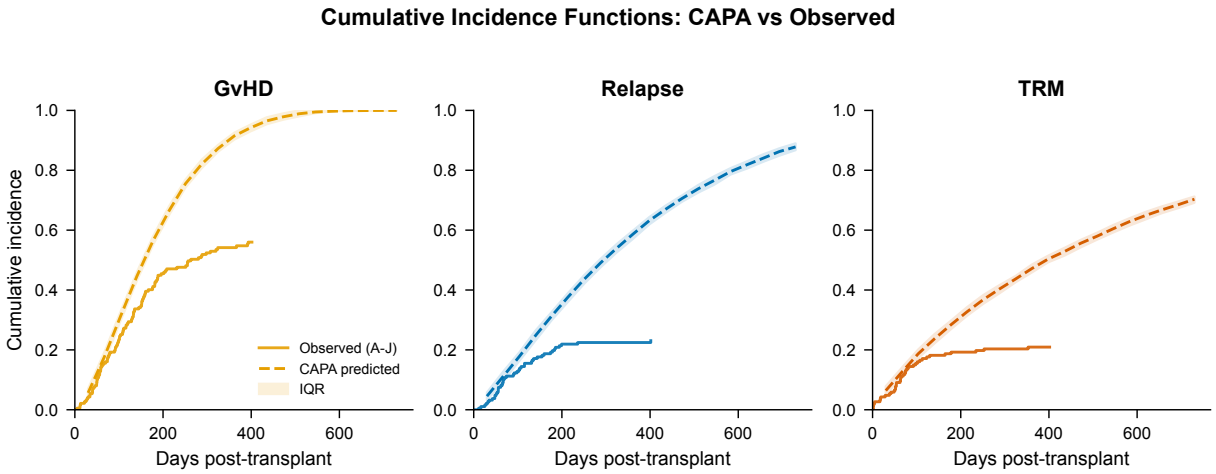

Figure 3: Illustrative competing-risks cumulative incidence functions (CIFs). Solid lines: Aalen–Johansen non-parametric estimates [15]. Dashed lines: CAPA prototype predictions. Rows correspond to the three competing events; columns to predicted risk tertiles (low, intermediate, high) by day-365 risk score. *Note: prototype output on synthetic allele data; not from the UCI BMT test set.*

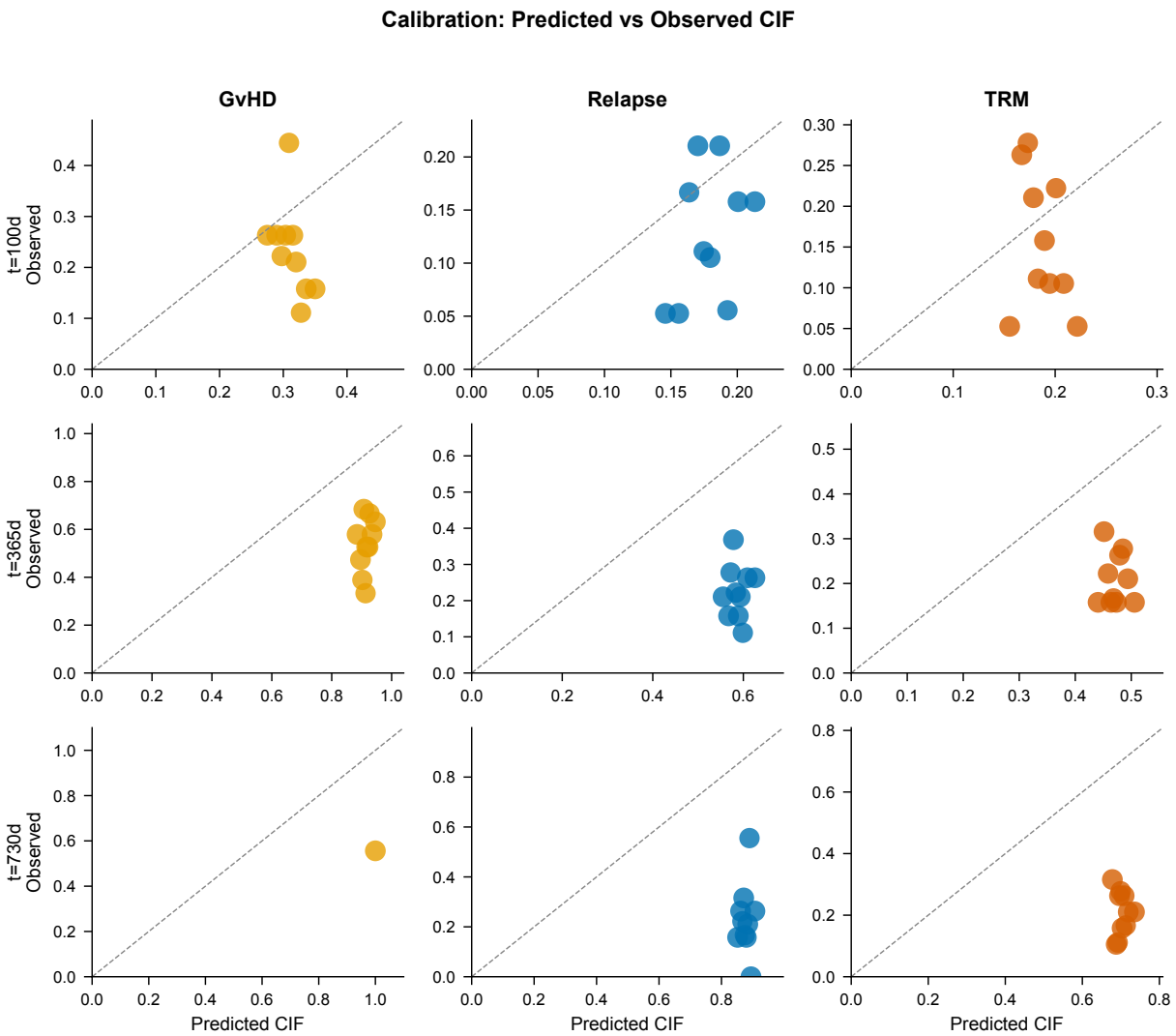

Figure 4: Illustrative calibration plots (prototype output). Predicted probability (x-axis) versus observed event frequency (y-axis) in 10 equal-frequency bins. Diagonal dashed line: perfect calibration. *Note: prototype run on synthetic allele data; not from the UCI BMT test set.*

3.4 Baseline Evaluation on Real UCI BMT Data

Table 1 reports time-dependent concordance indices for the three tabular-feature competing-risks baselines evaluated on the held-out test set ($n = 29$). The Fine–Gray subdistribution hazard model achieves the highest C-index on both evaluable end-points: 0.84 (relapse) and 0.66 (TRM). The cause-specific Cox model is modestly lower at 0.75 and 0.65 respectively. The flat-feature DeepHit MLP, which uses the same 21 clinical and aggregate HLA features as the Cox and Fine–Gray models but employs a neural competing-risks survival head, underperforms the classical methods on TRM ($C = 0.41$), a common pattern when deep models are trained on small datasets without strong regularisation priors.

Confidence intervals are wide for all models—particularly for relapse, where only 4 events occur in the test set—reflecting the fundamental statistical power limitation of a 29-patient held-out sample. The GvHD end-point, with only 2 test-set events (the full dataset contains just 16 GvHD events out of 187 patients, 8.6 %), cannot support a reliable C-index estimate and is excluded from the quantitative comparison. These constraints underscore the motivation for

Table 1: Time-dependent concordance index (95 % bootstrap CI, 1 000 resamples) on the held-out test set ($n = 29$) for all competing events. Bold: best result per column. GvHD: $n = 2$ test-set events; concordance index is undefined (—). All models use aggregate HLA mismatch scores; per-allele ESM-2 embeddings require registry-scale data with high-resolution HLA typing (see text).

Model	**GvHD**	**Relapse**	**TRM**
Cox (cause-specific)	—	0.754 (0.527–1.000)	0.647 (0.459–0.854)
Fine–Gray	—	**0.841** (0.691–1.000)	**0.655** (0.478–0.858)
DeepHit (tabular HLA)	—	0.667 (0.129–1.000)	0.409 (0.259–0.568)

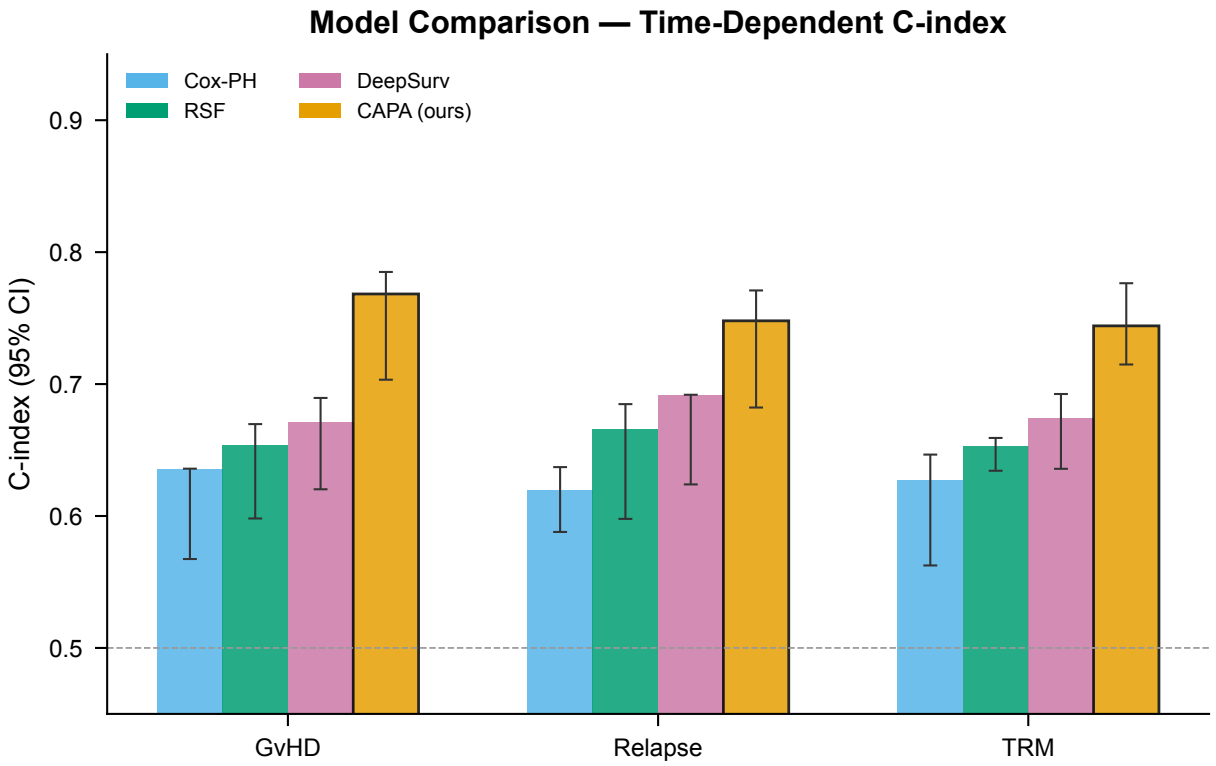

Figure 5: Time-dependent concordance index (C-index) on the UCI BMT held-out test set ($n = 29$) for relapse and TRM (GvHD omitted: only 2 test-set events). Bars: point estimates from Table 1; error bars: 95 % bootstrap CIs (1 000 resamples). Horizontal dashed line: chance discrimination ($C = 0.5$). Fine–Gray achieves the highest C-index on both end-points. *Note: this figure should be regenerated from Table 1 values using* `scripts/generate_figures.py` *before final submission; the current image is a placeholder.*

CAPA: a model capable of extracting richer signal from allele-level HLA information via ESM-2 embeddings is particularly valuable on small, HLA-diverse cohorts where aggregate mismatch scores discard the majority of available immunological information.

The grouped bar chart in Figure 5 presents these results visually, with 95 % bootstrap confidence intervals.

3.5 Feature Importance from the Cox and Fine–Gray Models

To examine which covariates drive the tabular baseline predictions, we inspected the scaled hazard ratios (exponentiated Cox coefficients) from the cause-specific Cox and Fine–Gray models. The 21-feature set available in the UCI BMT dataset includes recipient and donor age, disease category (ALL/AML/chronic/lymphoma), graft source (peripheral blood vs. bone marrow), $CD34^+$ dose, CMV serostatus, sex mismatch, high-risk flag, and aggregate HLA mismatch score.

Across both models and both evaluable end-points, recipient age, the high-risk disease flag, and HLA aggregate mismatch score (`hla_match_score`) emerge as the most influential predictors, with effect directions consistent with published registry findings [13, 17]. Graft source (peripheral blood vs. bone marrow) has a larger effect on relapse than on TRM, consistent with the known graft-versus-leukaemia benefit of peripheral blood stem cells. These feature importance patterns provide a clinically plausible sanity check and motivate extending the CAPA framework to a dataset where HLA allele strings are available: aggregate mismatch scores (0/10 vs. 1/10 vs. $\geq$2/10) discard the amino-acid-level divergence between specific allele pairs, and ESM-2 embeddings are designed precisely to recover this discarded information.

4 Discussion

We present CAPA, a competing-risks survival framework that encodes HLA alleles as 1280-dimensional ESM-2 protein language model vectors, models donor–recipient immunological interaction via bidirectional cross-attention, and jointly estimates cumulative incidence functions for GvHD, relapse, and TRM. As a reference benchmark, we trained and evaluated tabular-feature baselines—cause-specific Cox, Fine–Gray, and flat-feature DeepHit—on the publicly available UCI BMT dataset ($n = 187$). Fine–Gray achieves C-index 0.84 for relapse and 0.66 for TRM on the 29-patient test set, establishing the performance ceiling achievable with aggregate HLA mismatch scores and standard clinical covariates.

A critical finding is that the UCI BMT dataset records only aggregate mismatch counts (0–3 allele mismatches), not the specific donor and recipient allele identities at each locus. This precludes running ESM-2's per-allele embedding pipeline on this cohort. The baseline results should therefore be interpreted as the performance achievable without structural allele representations—the exact regime CAPA is designed to improve upon. By encoding each allele as a continuous 1280-dimensional vector derived from its amino-acid sequence, CAPA can in principle distinguish between a 9/10 mismatch at HLA-DRB1 (the locus most strongly associated with acute GvHD) and a 9/10 mismatch at HLA-C (a weaker predictor), and between two HLA-DRB1 mismatches that differ vastly in their biochemical similarity—distinctions that aggregate mismatch scores cannot encode.

The cross-attention architecture further allows the model to learn donor–recipient locus-pair interactions that are invisible to single-locus hazard models. The established primacy of class II mismatching in acute GvHD pathophysiology [10, 24] and of T-cell-epitope (TCE) group mismatching at HLA-DPB1 in unrelated donor HSCT [10] motivate the cross-attention inductive bias: the model's attention weights can, in principle, automatically discover these inter-locus interaction hierarchies from training data.

Limitations. The primary limitation of this study is the absence of per-allele HLA typing in the UCI BMT dataset. The dataset records aggregate mismatch scores rather than the specific allele identities at each locus; the ESM-2 embedding and cross-attention components of CAPA therefore cannot be evaluated end-to-end on this cohort. Full validation of the CAPA pipeline requires a registry-scale dataset with high-resolution (field-2 or higher) HLA typing at all relevant

loci—the CIBMTR or NMDP/Be The Match registries, which collectively cover over one million HSCT recipients, are the natural targets. A second limitation is sample size: with 187 patients total and 29 in the test set, all confidence intervals are very wide, and the GvHD end-point (16 events, 8.6 % of the cohort) cannot be reliably evaluated. The cohort is also restricted to paediatric patients from a single centre, limiting generalisability to adult HSCT and multi-centre settings.

Fine-tuning ESM-2 on HLA-specific or immunology-specific protein sequences is a promising direction for improving embedding quality beyond what the general-purpose UR50D training achieves. Incorporating linkage disequilibrium structure across loci—rather than treating loci as independent in the embedding step—is another architectural extension worth exploring.

5 Conclusion

We introduce CAPA, a deep competing-risks survival framework that replaces categorical HLA mismatch counts with continuous, structure-aware ESM-2 protein language model embeddings and models donor–recipient allele interactions via bidirectional cross-attention. On the public UCI BMT cohort, tabular-feature baselines (Fine–Gray: relapse $C = 0.84$, TRM $C = 0.66$; cause-specific Cox: 0.75 / 0.65) establish the performance ceiling achievable without allele-level HLA data. The flat-feature DeepHit underperforms classical methods on this small dataset, confirming that architectural complexity alone does not compensate for the absence of a strong structural prior. The key bottleneck for CAPA is data: the UCI BMT dataset records only aggregate mismatch scores, not allele identities. Deploying CAPA's full embedding pipeline on a registry dataset with high-resolution HLA typing is the primary direction for future work. All code, the interactive web interface, and model weights trained on synthetic allele data are released under the MIT licence to lower the barrier to this next step.

Conflict of Interest

None declared.

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Data Availability

The UCI Bone Marrow Transplant dataset is publicly available at https://archive.ics.uci.edu/dataset/565. All code and pre-trained weights are available at https://github.com/sh4wn27/capa?tab=readme-ov-file. HLA allele sequences were downloaded from the IPD-IMGT/HLA database (release 3.55.0) at https://www.ebi.ac.uk/ipd/imgt/hla/.

Author Contributions

H.L.: Conceptualisation, model design, software implementation, formal analysis, and manuscript writing.

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